Money Wiz CRM Project

1. Project Overview

A Python-based CRM and Sales Visualization system designed for retail businesses. It enables **Salespersons** to manage customer data and analyse personal sales, while **Region Managers** can visualize store and region-level metrics for data-driven decisions.

This project is based on targeting two types of business users in majority of retail stores – Salesperson and Managers (Region Manager).

Salesperson – Responsible for entering and maintaining customer records and reviewing individual sales performance.

Region Managers – Uses system-generated visualizations to compare sales across regions, stores, and individuals and to monitor product return patterns.

The project demonstrates practical applications of **pandas**, **matplotlib**, and **Excel I/O** within a CRM context. It integrates CRUD operations, data validation, and analytics visualizations in one unified workflow.

---

2. Business Use Case

Salespersons often rely on spreadsheets to maintain scattered customer details. MoneyWiz standardizes this process by providing a structured interface to:

- Add and update customer information
- Store product purchase details
- Track past sales through visual summaries

Region Managers need a high-level view to identify trends and performance gaps across stores and regions. The system enables managers to:

- Compare sales across regions and stores
- Identify high-performing salespersons
- Monitor top-selling products
- Analyse item returns store-wise
- Support incentive, inventory, and promotion decisions

The tool supports data-driven decision-making for both operational and tactical business needs.

3. System Architecture

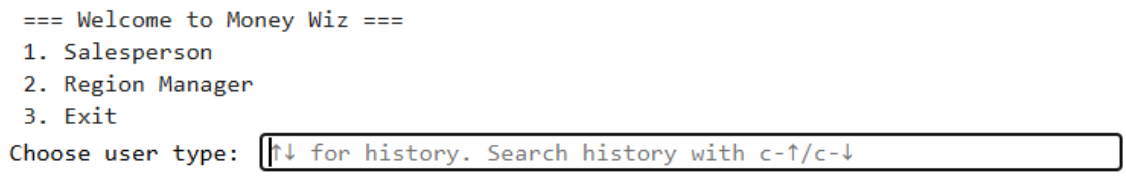
3.1. Technology Stack

- **Python**
- **Pandas** (data manipulation)
- **Matplotlib** (visualizations)
- **Excel (.xlsx)** file-based storage

3.2. Core Modules

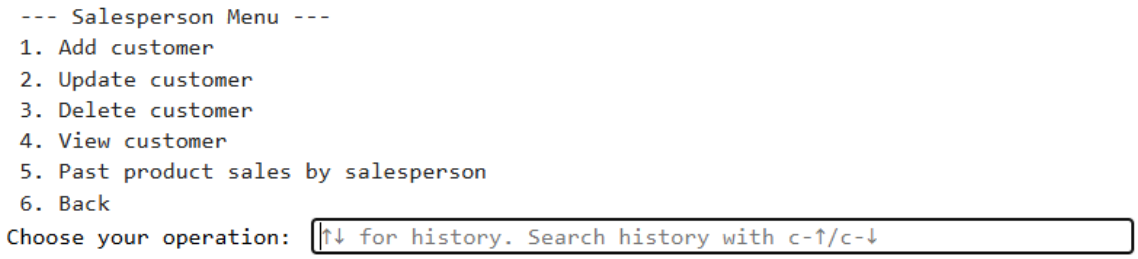
3.2.1. Main menu –

The main menu which acts as the entry point of the system. Display which type of users can access the system.



3.2.2. Salesperson menu –

This menu allows salespersons to perform CRUD operations (Create, Read, Update, Delete) for customer operations and personal sales visualization. The user must enter 1 in the Main menu to access the salesperson menu.



Salesperson Operations -

3.2.2.1. Add customer operation –

- This operation allows new customer details entry to the existing excel file.
- The user must type 1 and add customer details in the required fields and once the details are entered, the customer details are added, and operation completion message gets displayed as shown below -

```
--- Salesperson Menu ---
1. Add customer
2. Update customer
3. Delete customer
4. View customer
5. Past product sales by salesperson
6. Back
Choose your operation:  1
customerName:  Aniket
customerType:  Retail
product:  Laptop
quantity:  2
region:  East
date:  14-11-2025
UnitPrice:  120
StoreLocation:  Store A
Discount:  0
Salesperson:  Eva
TotalPrice:  240
PaymentMethod:  Online
Promotion:
Returned:  0
OrderID:  REG101520
ShippingCost:  10
OrderDate:  10-11-2025
DeliveryDate:  13-11-2025
RegionManager:  Eric
Customer added successfully.
```

3.2.2.2. Update customer operations –

- This operation allows updating of data of existing customer in the system dataset.
- The user must type 2 and enter the name of the customer whose details are to be updated, only then the details can be entered in any field which needs to be updated and can leave rest of the fields blank.
- Once the data entry is completed, the customer details get updated in the system and the operation completion message gets displayed as shown below -

```
--- Salesperson Menu ---
1. Add customer
2. Update customer
3. Delete customer
4. View customer
5. Past product sales by salesperson
6. Back
Choose your operation:  2
Enter customer name to update:  Aniket
New customerType:  Wholesale
New product:
New quantity:
New region:
New date:
New UnitPrice:
New StoreLocation:
New Discount:
New Salesperson:
New TotalPrice:
New PaymentMethod:
New Promotion:
New Returned:
New OrderID:
New ShippingCost:
New OrderDate:
New DeliveryDate:
New RegionManager:
Customer updated successfully.
```

Non existing customer – if the customer details are not present in the dataset, then the message – "Customer not found!" gets displayed as shown below -

```
--- Salesperson Menu ---
1. Add customer
2. Update customer
3. Delete customer
4. View customer
5. Past product sales by salesperson
6. Back
Choose your operation:  2
Enter customer name to update:  Ronaldo
Customer not found!
```

3.2.2.3. Delete customer operations –

- This operations allows the user to remove details of exiting customer from the system dataset.

- The user must type 3 and enter the customer name and then once, that customer details are removed, the message – “Customer deleted successfully.” is displayed as shown below -

```
--- Salesperson Menu ---
1. Add customer
2. Update customer
3. Delete customer
4. View customer
5. Past product sales by salesperson
6. Back
Choose your operation:  3
Enter customer name to delete:  Aniket
Customer deleted successfully.
```

Delete non-existing customer – if the customer details are not present in the dataset, then the message – “Customer not found!” gets displayed as shown below -

```
--- Salesperson Menu ---
1. Add customer
2. Update customer
3. Delete customer
4. View customer
5. Past product sales by salesperson
6. Back
Choose your operation:  3
Enter customer name to delete:  Aniket
Customer not found!
```

3.2.2.4. View customer operations –

- This operation allows the user to view details of existing customer from the system dataset.
- The use must type 4 and enter name of the customer who details are to be displayed and once the data gets fetched, all the customer details are displayed as shown below -

```
--- Salesperson Menu ---
1. Add customer
2. Update customer
3. Delete customer
4. View customer
5. Past product sales by salesperson
6. Back

Choose your operation:  4
Enter customer name to view:  Alex Allen

Customer details:
           Date Region Product  Quantity  UnitPrice StoreLocation  \
0    23/02/2023   East  Laptop        14     163.60       Store C
229  05/10/2024   East  Tablet        20     576.30       Store C
359  25/02/2025  North  Tablet        13     238.34       Store C

    CustomerType  Discount Salesperson  TotalPrice PaymentMethod Promotion  \
0      Wholesale      0.73         Eva    2290.400        Online  FREESHIP
229    Wholesale      0.15         Eva   10949.700   Credit Card       NaN
359       Retail      4.43      Carlos    2788.578   Credit Card    SAVE10

     Returned    OrderID CustomerName  ShippingCost   OrderDate DeliveryDate  \
0           0  REG100000   Alex Allen         43.34  23/02/2023   27/02/2023
229         0  REG100229   Alex Allen         25.22  05/10/2024   09/10/2024
359         1  REG100359   Alex Allen          7.90  25/02/2025   06/03/2025

    RegionManager
0            Eric
229          Eric
359          Ryan
```

Viewing non-existing customer – if the customer details are not present in the dataset, then the message – "Customer not found!" gets displayed as shown below -

```
--- Salesperson Menu ---
1. Add customer
2. Update customer
3. Delete customer
4. View customer
5. Past product sales by salesperson
6. Back

Choose your operation:  4
Enter customer name to view:  Ronaldo

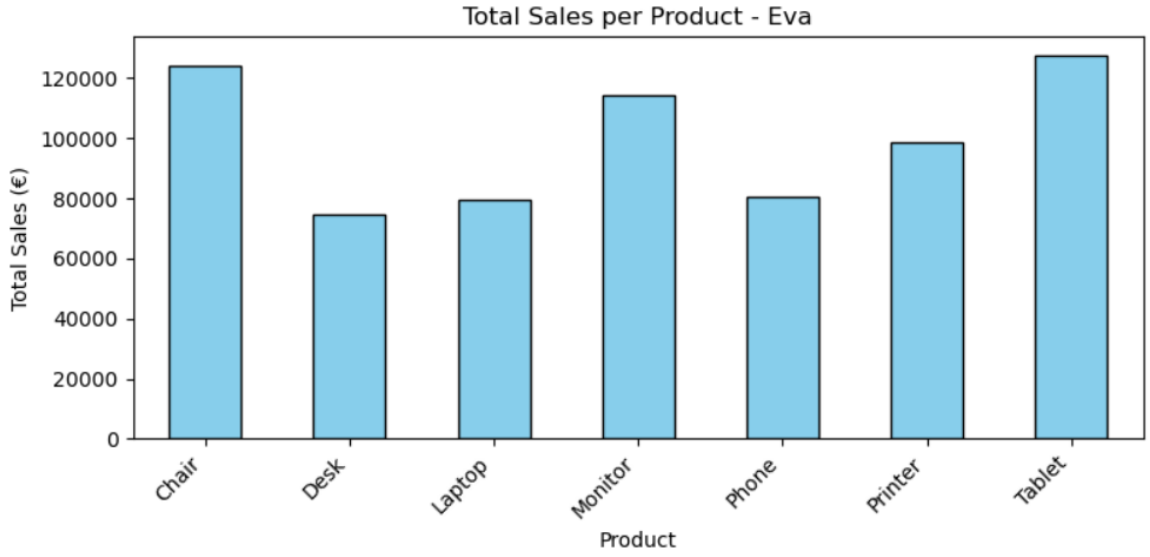
Customer not found!
```

3.2.2.5. Viewing past sales for individual salesperson –

- This operation allows the user to view the salesperson past product sales.
- The user must enter operation 5 and the name of the salesperson whose sales data is to be display, then the sales summary of that salesperson gets displayed along with graphical visualization of their sales data as shown below -

```
--- Salesperson Menu ---
1. Add customer
2. Update customer
3. Delete customer
4. View customer
5. Past product sales by salesperson
6. Back
Choose your operation:  5
Enter salesperson name:  Eva

Sales Summary for Eva:
  Chair: €124112.16
  Desk: €74814.89
  Laptop: €79446.04
  Monitor: €114360.56
  Phone: €80515.92
  Printer: €98766.25
  Tablet: €127343.76
```



3.2.3. Region Manager menu –

This menu allows Region Managers to view visualization of sales data of many different business metrics for analysis and data-driven decision making. The user must enter 2 in the main menu to access Region Manager menu.

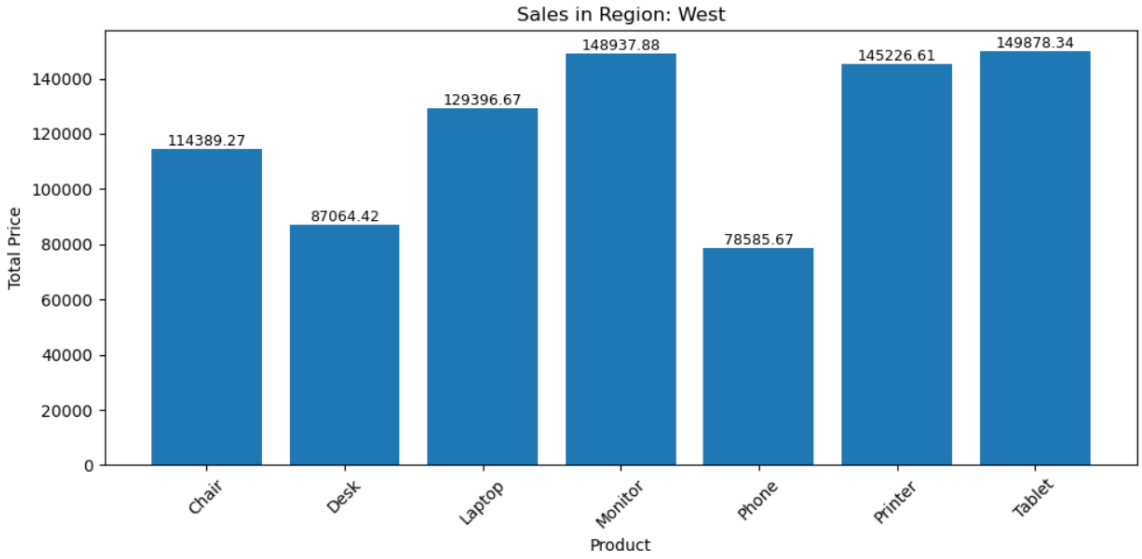
```
---Region Manager Menu ---
1. Region-wise sale
2. Store-wise sale
3. Person-wise sale
4. max product sales by store
5. salesperson max sales
6. store wise return items
7. Back
Choose your operation:  ↑↓ for history. Search history with c-↑/c-↓
```

Region Manager Operations -

3.2.3.1. Region-wise sale graph –

- This operation allows the Manager to view sales data from each region in visual form.
- The user must type 1 to select this operation in Region Manager menu and enter the name of the region for which data is to be displayed, then the sales data is displayed in graphical visualization as shown below -

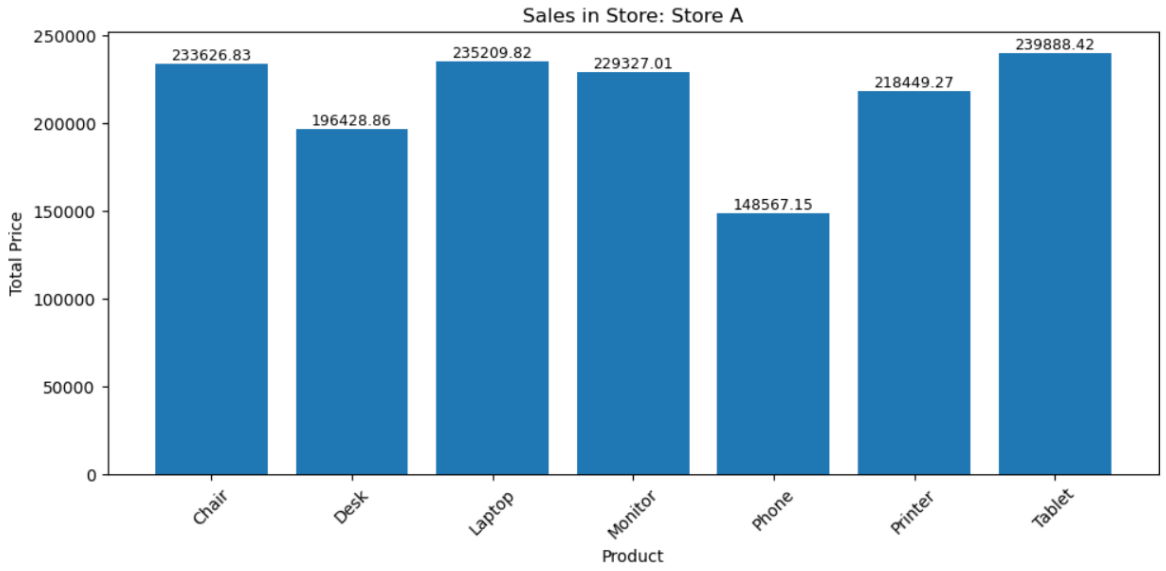
```
---Region Manager Menu ---
1. Region-wise sale
2. Store-wise sale
3. Person-wise sale
4. max product sales by store
5. salesperson max sales
6. store wise return items
7. Back
Choose your operation:  1
Enter region name:  West
```



3.2.3.2. Store-wise sale –

- This operation allows the Manager to view sales data from each store in visual form.
- The user must type 2 to select this operation in Region Manager menu and enter the name of the store for which data is to be displayed, then the sales data is displayed in graphical visualization as shown below -

```
---Region Manager Menu ---
1. Region-wise sale
2. Store-wise sale
3. Person-wise sale
4. max product sales by store
5. salesperson max sales
6. store wise return items
7. Back
Choose your operation:  2
Enter store name:  Store A
```

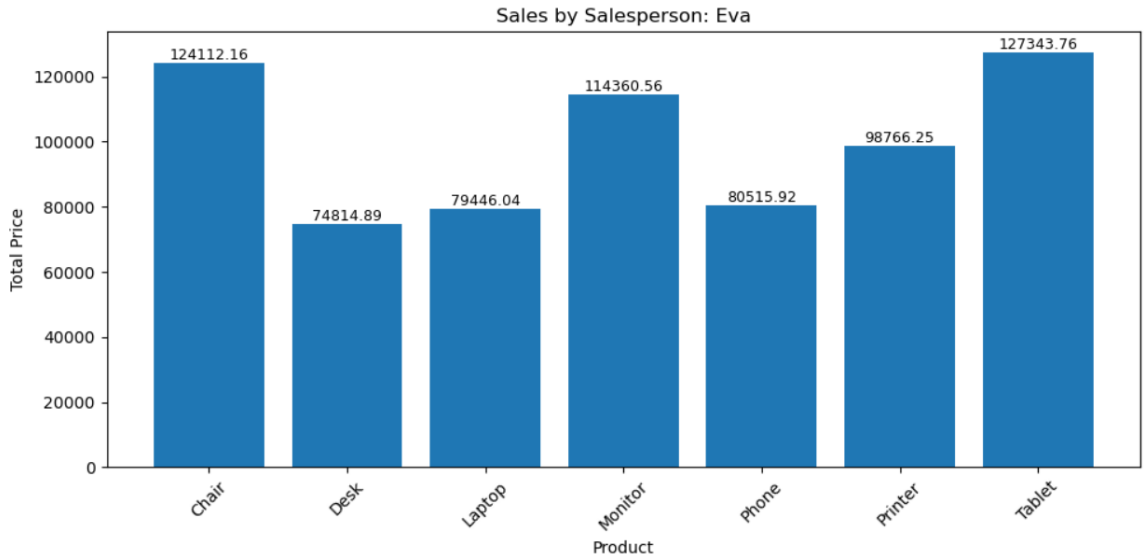


3.2.3.3. Person-wise sales –

- This operation allows the Manager to view sales data from each salesperson in visual form.
- The user must type 3 to select this operation in Region Manager menu and enter the name of the salesperson for whom data is to be displayed, then the sales data is displayed in graphical visualization as shown below -

```
---Region Manager Menu ---
1. Region-wise sale
2. Store-wise sale
3. Person-wise sale
4. max product sales by store
5. salesperson max sales
6. store wise return items
7. Back

Choose your operation:  3
Enter salesperson name:  Eva
```

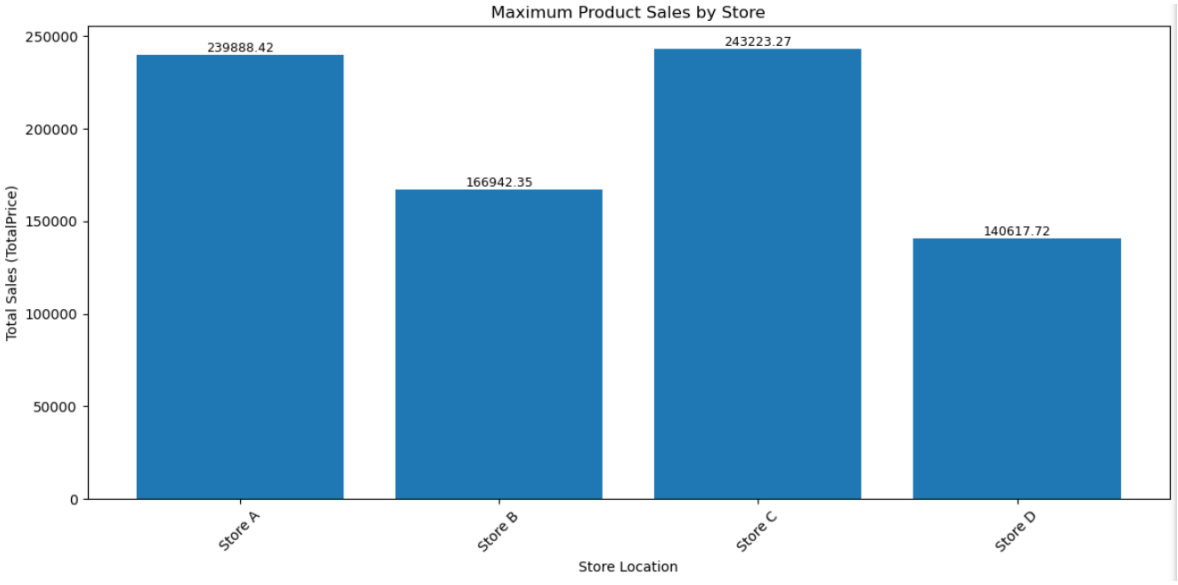


3.2.3.4. Max product sales by store –

- This operation allows the Manager to view maximum product sales by each store in visual form.
- The user must type 4 to select this operation in Region Manager, then the sales data is displayed in graphical visualization as shown below -

```
---Region Manager Menu ---
1. Region-wise sale
2. Store-wise sale
3. Person-wise sale
4. max product sales by store
5. salesperson max sales
6. store wise return items
7. Back
Choose your operation:  4

=== Max Product Sales by Store ===
   StoreLocation  Product   TotalPrice
6        Store A   Tablet  239888.4205
12       Store B  Printer  166942.3545
19       Store C  Printer  243223.2665
24       Store D  Monitor  140617.7245
```

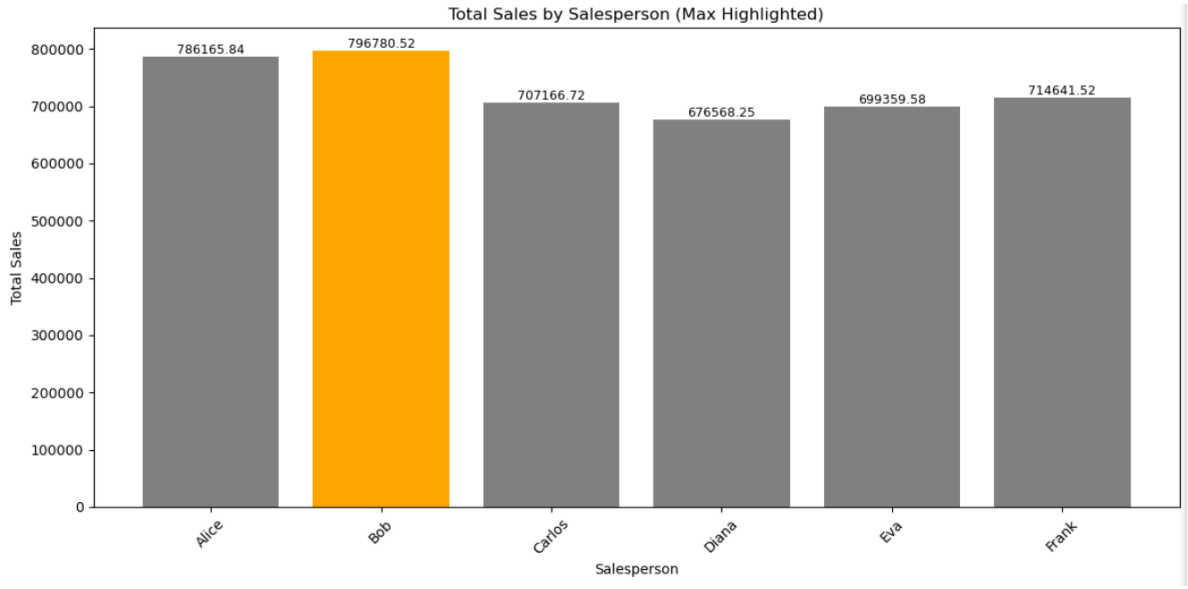


3.2.3.5. salesperson max sales –

- This operation allows the Manager to view sales data for the salesperson with maximum sales in visual form.
- The user must type 3 to select this operation in Region Manager menu, then the sales data is displayed in graphical visualization as shown below, the salesperson with maximum sales gets highlighted in yellow colour in the sales graph -

```
---Region Manager Menu ---
1. Region-wise sale
2. Store-wise sale
3. Person-wise sale
4. max product sales by store
5. salesperson max sales
6. store wise return items
7. Back

Choose your operation:  5
```

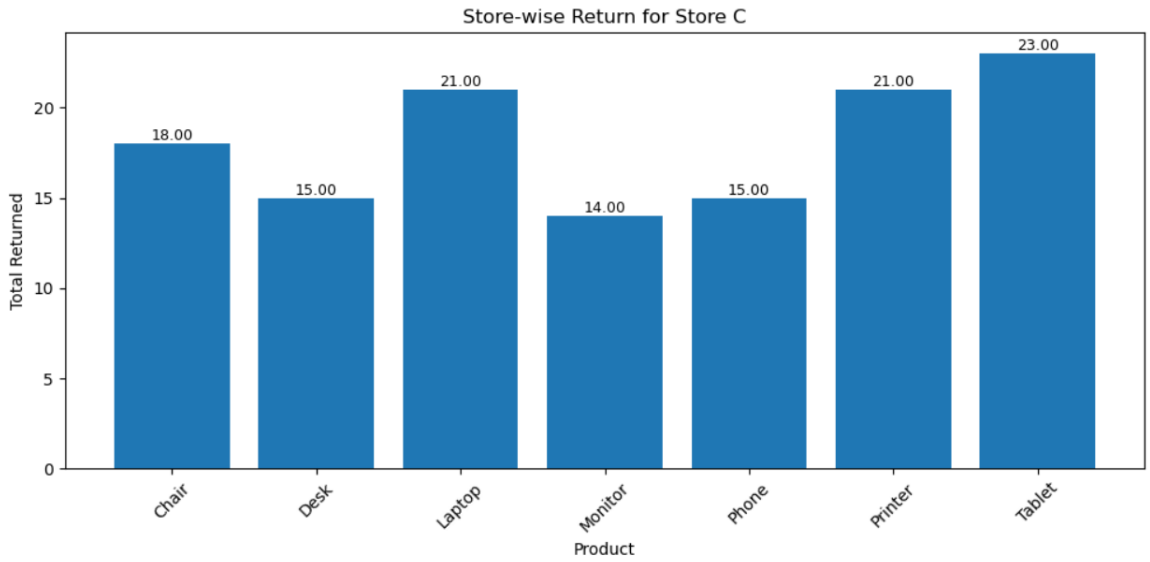


3.2.3.6. store wise return items –

- This operation allows the Manager to view number of returned items for specific store in visual form.
- The user must type 3 to select this operation in Region Manager menu and enter the name of the store for whom returned items data is to be displayed, then the sales data is displayed in graphical visualization as shown below -

```
---Region Manager Menu ---
1. Region-wise sale
2. Store-wise sale
3. Person-wise sale
4. max product sales by store
5. salesperson max sales
6. store wise return items
7. Back

Choose your operation:  6
Enter store name:  Store C
```



---

6. Strengths of the Project

- Fully menu-driven experience suitable for beginners
- Clean separation of user roles
- Intuitive CRUD operations for salespersons
- Insightful analytics for managers
- Strong use of pandas grouping and aggregation
- Matplotlib visualizations tailored for business insights
- Lightweight, Excel-based backend accessible to non-technical users

---

7. Limitations and Improvement Opportunities

- Data stored in Excel limits scalability
- Manual input may cause data inconsistencies
- Lacks authentication for user roles
- No GUI (Tkinter/Streamlit can enhance UX)
- Visualization dashboard can be moved to Power BI or Tableau

Future enhancements could include a **database migration**, **web dashboard**, and **forecasting models** for predictive insights.

---

8. GitHub Repository

Full project code, dataset, and documentation are available at:

https://github.com/vighnesh-shetty-vs/MoneyWiz-CRM-Project